Moment Generating functions

Moment Generating Function of a random variable X (discrete or continuous) is defined as $E(e^{tX})$ where t is a real variable, denoted by $M(t)$

$$M(t) = \begin{cases} \sum_r e^{t x_r} p(X = x_r) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx & \text{if } X \text{ is continuous.} \end{cases}$$

Properties:— (i) $M(t) = \sum_{n=0}^{\infty} \frac{t^n E(X^n)}{n!}$

So, $E(X^n)$ is the co-efficient of $\frac{t^n}{n!}$ in the expansion of $M(t)$ in series power of t.

(2) $\mu_n' = E(X^n) = \left[\frac{d^n}{dt^n} M(t)\right]_{t=0}$

(3) If $Y = aX + b$, then $M_Y(t) = e^{bt} M_X(at)$

(4) If X and Y are independent random variables and $Z = X + Y$ then $M_Z(t) = M_X(t) \cdot M_Y(t)$.

Pb: 1 If a RV has the moment generating function

$$M(t) = \frac{pe^t}{1-qe^t}$$

Find $E(X)$.

Soln:-

we know, $E(X) = \left[\frac{d}{dt} M(t)\right]_{t=0}$

Here $\frac{d}{dt} M(t) = \frac{d}{dt}\left(\frac{pe^t}{1-qe^t}\right) = \frac{(1-qe^t)pe^t - pe^t(-qe^t)}{(1-qe^t)^2}$

$$= \frac{pe^t}{(1-qe^t)^2}$$

$$E(X) = \frac{p}{(1-q)^2}$$

Pb: 2 If X represents the outcome, when a fair die is tossed, find the MGF of X and hence find $E(X)$, $Var(X)$.

Soln:- A probability mass function for the event of throughing a die

X	1	2	3	4	5	6
$p(X)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

$$M(t) = \sum_i e^{t x_i} p(X=x_i) = e^{t.1} p(X=1) + e^{t.2} p(X=2) + e^{t.3} p(X=3) + e^{t.4} p(X=4) + e^{t.5} p(X=5) + e^{t.6} p(X=6)$$

$$M(t) = e^t \cdot \frac{1}{6} + e^{2t} \cdot \frac{1}{6} + e^{3t} \cdot \frac{1}{6} + e^{4t} \cdot \frac{1}{6} + e^{5t} \cdot \frac{1}{6} + e^{6t} \cdot 1/6$$

$$= \frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)$$

$$E(X) = \left[\frac{d}{dt} M(t)\right]_{t=0} = \left[\frac{1}{6}\left(e^t + 2e^{2t} + 3e^{3t} + 4e^{4t} + 5e^{5t} + 6e^{6t}\right)\right]_{t=0}$$

$$= \frac{1}{6}[1+2+3+4+5+6] = 7/2$$

$$E(X^2) = \left[\frac{d^2}{dt^2} M(t)\right]_{t=0} = \left[\frac{1}{6}\left(e^t + 4e^{2t} + 9e^{3t} + 16e^{4t} + 25e^{5t} + 36e^{6t}\right)\right]_{t=0}$$

$$= \left[\frac{1}{6}\left(1 + 4 + 9 + 16 + 25 + 36\right)\right] = \frac{91}{6}$$

$$\therefore Var(X) = E(X^2) - [E(X)]^2$$

$$= \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

$$\text{standard deviation} = \sqrt{Var(X)} = \sqrt{\frac{35}{12}}$$

Let X be a discrete random variable has the probability mass function

$$P(X=r) = {}^nC_r \, p^r q^{n-r}, \quad r = 0, 1, 2, \ldots n$$
$$p+q=1$$

Find $M(t)$, $E(X)$, $Var(X)$.

Soln:-

$$M(t) = \sum_{r=0}^{n} e^{tr} P(X=r) = \sum_{r=0}^{n} e^{tr} \, {}^nC_r \, p^r q^{n-r}$$

$$= \sum_{r=0}^{n} {}^nC_r (pe^t)^r q^{n-r} = (pe^t+q)^n$$

$$\left[(a+b)^n = \sum_{r=0}^{n} {}^nC_r \, a^r b^{n-r}\right]$$

$$E(X) = \left[\frac{d}{dt} M(t)\right]_{t=0} = \left[n(pe^t+q)^{n-1} \cdot pe^t\right]_{t=0}$$

$$= n(p+q)^{n-1} \cdot p \cdot 1 = np \qquad [\text{because } p+q=1]$$

$$E(X^2) = \left[\frac{d^2}{dt^2} M(t)\right]_{t=0} = \left\{\frac{d}{dt}\left[n(pe^t+q)^{n-1} pe^t\right]\right\}_{t=0}$$

$$= \left[n(n-1)(pe^t+q)^{n-2} \cdot pe^t \cdot pe^t + n \cdot (pe^t+q)^{n-1} \cdot pe^t\right]_{t=0}$$

$$= n(n-1)p^2 + np = np\left[(n-1)p+1\right]$$

$$Var(X) = E(X^2) - [E(X)]^2 = np\left[(n-1)p+1\right] - n^2p^2$$

$$= np(n-1)p + np - n^2p^2 = n^2p^2 - np^2 + np - n^2p^2$$

$$= np - np^2 = np(1-p) = npq$$